ChatGPT

Minimizing total treatment time

- We minimize $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$. This is a linear program, so the optimum lies at an extreme point of the feasible set [1] (indeed a linear objective over a convex polytope has a global minimum at a vertex [2]).
- The only feasible vertices are $(x_1, x_2) = (1, 0)$ (giving $T = 1$) and $(0, 1/a_2)$ (giving $T = 1/a_2$). We compare $T = 1$ vs. $T = 1/a_2$.
- The given code shows a_2 is just above 1 (it tends to 1 from above). Hence $1/a_2 < 1$, so the smaller total time is at $(0,\ 1/a_2)$. Thus the minimiser is $x_1 = 0$, $x_2 = 1/a_2$. Numerically $1/a_2 \approx 1.00$ (to two-decimal accuracy), so
$x_1 \approx 0.00,\ x_2 \approx 1.00.$

These values give the minimum time (since using the faster second therapy exclusively yields $T = 1/a_2$, which is slightly below 1) [1].

Answer: $x_1 \approx 0.00,\ x_2 \approx 1.00$ (to two decimal places), with justification as above [1] [2].

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[1] [2] Linear programming - Wikipedia
https://en.wikipedia.org/wiki/Linear_programming